AKASH

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ABOUT

- Agricultural Engineering graduate with concentrated research experience in irrigation water quality assessment, crop-water modeling, and the water-climate nexus.
- Skilled in field experimentation, lab work, process-based simulation and statistical analysis.
- Motivated to acquire advanced knowledge in machine learning and contribute to innovative water management solutions and groundwater monitoring to support environmental sustainability.

EDUCATION

- **Master of Science in Irrigation and Water Management** (Expected March 2027)
  Department of Irrigation and Water Management
  Bangladesh Agricultural University, Mymensingh, Bangladesh
  Current CGPA: **3.28 out of 4.00** (Thesis Sesmeter ongoing)
- **Bachelor of Science in Agricultural Engineering** (October 2025)
  Faculty of Agricultural Engineering and Technology
  Bangladesh Agricultural University, Mymensingh, Bangladesh
  CGPA: **3.302 out of 4.00**
- **Higher Secondary Certificate** (July 2019)
  Science group
  Rajuk Uttara Model College, Dhaka, Bangladesh
  GPA: **5.00 out of 5.00**

RESEARCH INTERESTS

GIS and Remote Sensing, Hydrological Modeling, Groundwater Management, Watershed Modeling, Machine Learning, Soil and Water Conservation, Agricultural Water Management, Climate Change Impacts on Water-Food Systems, Wastewater Quality Assessment and Environmental Impacts.

RESEARCH EXPERIENCE

Position: Research Assistant (June 2022–Present)

Department of Irrigation and Water Management, Bangladesh Agricultural University

Supervisor & Co-Supervisor: Dr. Md. Touhidul Islam & Dr. A.K.M. Adham

- Designing experimental layouts, assembling materials, and monitoring progress in the fields and laboratories
- Collecting and analyzing research data; creating flowcharts, figures, and tables
- Preparing reports, presentations, and posters for conferences or seminars
- Writing and assisting in preparing thesis manuscripts and research papers

INTERNSHIP EXPERIENCE

- Alim Industries Limited: Observed the manufacturing and assembly processes of power tillers and combine harvesters.
- Bangladesh Agricultural Development Corporation (BADC): Gained hands-on exposure to deep tubewell and minor irrigation infrastructure.
- Bangladesh Institute of Nuclear Agriculture (BINA): Gained exposure to solar dryers for agro-produce, mechanical rice transplanting techniques, and IoT-based smart agro-farm security systems.

PUBLICATIONS

1. Islam, M. T., Akash, Khatun, M. R., Jahan, N., Hakim, A., Saha, A. K., Hossain, S. A. A. M., Rahman, M. M., & Adham, A. K. M. (2026). Future-proofing raised bed irrigation for rabi maize using sensor-based soil moisture monitoring, AquaCrop modeling, and CMIP6-based climate projections in Bangladesh. Agricultural Water Management, 335, 110740. https://doi.org/10.1016/j.agwat.2026.110740

2. Islam, M. T., **Akash**, Khatun, M. R., Jahan, N., Islam, M. R., Partho, D., Kibria, M., & Adham, A. (2024). Regional irrigation

water quality index for the Old Brahmaputra River, Bangladesh: A multivariate and GIS-based spatiotemporal assessment. Results in Engineering, 24, 103667. https://doi.org/10.1016/j.rineng.2024.103667

3. Islam, M. T., Shalehin, M., Jahan, N., Islam, M. R., **Akash**, & Adham, A. K. (2024). Modeling Boro rice water requirements and irrigation schedules in Mymensingh, Bangladesh, under subtropical climate change. Results in Engineering, 24, 103665. https://doi.org/10.1016/j.rineng.2024.103665

TECHNICAL AND RESEARCH SKILLS

- **Academic Writing, Scientific Communication, and Data Analysis**
  - Microsoft Office (Word, Excel, and PowerPoint), SPSS, OriginPro, Minitab
- **Water Flow, Solute Transport, and Crop Water Requirement Modeling**
  - DSSAT, AquaCrop, CROPWAT, and HYDRUS
- **Programming**
  - Python, MATLAB, C, Java, CSS
- **Computer-Aided Design & Visualization**
  - AutoCAD, Draw.io and SketchUp
- **Geospatial Analysis and Mapping**
  - ArcGIS
- **Water Resources and Hydrological Modeling**
  - MODFLOW, SWAT, and HEC-HMS
- **Field Experimentation and Lab Work**
  - Experimental design, field-plot preparation, instrument installation and monitoring, crop and field-site monitoring, soil and water sampling, soil-moisture analysis, laboratory analysis, data collection, model calibration and validation.

PROJECT

Climate-Smart Irrigation and Nitrogen Optimization for Sustainable Boro Rice: Integrating AquaCrop Modeling with CMIP6 Projections in Bangladesh

Conducted a two-year (2024–2025) field experiment on Boro rice (BRRI Dhan 88) to assess sustainability under future climate conditions, using three irrigation strategies (CF, AWD-15, AWD-20) and three nitrogen levels (90, 67.5, 45 kg N/ha). A Research paper based on this project has been submitted is under review.

LANGUAGE PROFICIENCY

- Bengali (Native)
- English (CEFR level B2: Upper-Intermediate)
  - **IELTS: Overall band score - 6** (Listening - **6**, Reading – **6.5**, Writing - **6**, Speaking – **5.5**)
- Hindi (Advanced Conversational Fluency)

CERTIFICATIONS

- Department of Agricultural Extension Education (Certificate of Participation)
- Alim Industries Limited (Certificate of Internship)
- Bangladesh Agricultural Development Corporation (Certificate of Completion)
- Agricultural Engineering and Technology (Internship Certificate)

REFERENCES

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